



Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a Research Case Study



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Acronyms used in this material (1/2)

- ▣ AAPL: Apple Inc. (Nasdaq ticker)
- ▣ ACM: Association for Computing Machinery
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ AUD: Australian Dollar (ISO 4217 code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ BPQ: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- ▣ CAD: Canadian Dollar (ISO 4217 code)
- ▣ CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DE: Deutschland (jurisdiction of the CC licence) (Germany (jurisdiction of the CC licence))
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: E-mini S&P 500 futures (CME ticker)
- ▣ EURUSD: Euro / US dollar exchange rate (FX pair code)
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GC: Gold futures, COMEX ticker
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)

Acronyms used in this material (2/2)

- ▣ KDD: Knowledge Discovery and Data Mining (ACM SIGKDD conference)
- ▣ LARS: Least-Angle RegreSsion (algorithm)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MCS: Model Confidence Set
- ▣ MSE: Mean Squared Error
- ▣ NG: Natural Gas futures, NYMEX ticker
- ▣ OLS: Ordinary Least Squares
- ▣ PDE: Partial Differential Equation
- ▣ PRDS: Positive Regression Dependence on a Subset (of p-values)
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SHAR: Semivariance HAR model
- ▣ TSLA: Tesla, Inc. (Nasdaq ticker)
- ▣ VOLARE: VOLatility Archive for Realized Estimates (database)
- ▣ XOM: Exxon Mobil Corporation (NYSE ticker)

Chapter Route and Outcomes

Route of this chapter (2×80 minutes)

- A research case study: the method of [Gu et al. \(2024\)](#), published for freight rates at Amazon, transferred to daily realised variance of 50 VOLARE assets [Cipollini et al. \(2026\)](#)
- Session 1: realised measures, the data, the HAR benchmarks, path signatures
- Session 2: the paper, a design fixed before the results, results, robustness, one pass of the AI-assisted discovery loop
- Implementation detail sits in the Appendix at the end

After this chapter you can

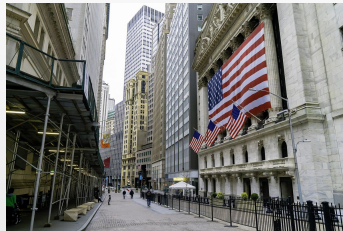
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets with DM tests, model confidence sets and Holm/BH corrections
- pre-register a method transfer and report a null result with its mechanism

Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
 - ▶ Tail-risk measures (Chapter 7), option prices (Chapter 12), portfolio weights (Chapter 4)
- Intraday data make daily variance almost observable: realised variance (RV, Chapter 9)
 - ▶ So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat: the HAR model [Corsi \(2009\)](#) (Chapter 9)
 - ▶ Three averages of past RV; linear; estimated by OLS in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

The Research Question

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
 - ▶ Signature features enter a sparse linear regression (two-step LASSO)
 - ▶ Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
 - ▶ Question: does the same recipe improve on HAR-type models for realised variance?
- Two separate ingredients to test: the signature features and the kernel weights



Wall Street and the New York Stock Exchange, 25
March 2020

Photo: [Billie Grace Ward \(2020\)](#); CC BY 2.0; [Wikimedia Commons](#)

Realised Variance and Quadratic Variation

- Log price p_t with $dp_t = \mu_t dt + \sigma_t dW_t + dJ_t$; day t split into M intraday returns $r_{t,i}$
 - ▶ $RV_t = \sum_{i=1}^M r_{t,i}^2 \xrightarrow{M \rightarrow \infty} \int_{t-1}^t \sigma_s^2 ds + \sum_{t-1 < s \leq t} (\Delta J_s)^2$ Barndorff-Nielsen & Shephard (2002)
- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance IV_t ; the error is mixed Normal
 - ▶ $\sqrt{M}(RV_t - IV_t) \Rightarrow MN(0, 2IQ_t)$, $IQ_t = \int_{t-1}^t \sigma_s^4 ds$ (integrated quarticity)
- Microstructure noise makes very fine sampling biased; five-minute returns are the standard compromise (Chapter 9)
- Chapter 9 derives these results; they are used here as given

The Realised Measures in VOLARE

- Semivariances [Barndorff-Nielsen, Kinnebrock & Shephard \(2010\)](#): $RV_t^+ = \sum_i r_{t,i}^2 1\{r_{t,i} > 0\}$,
 $RV_t^- = \sum_i r_{t,i}^2 1\{r_{t,i} < 0\}$
 - ▶ They add up: $RV_t = RV_t^+ + RV_t^-$ (holds in VOLARE to machine precision)
- Realised quarticity: $RQ_t = \frac{M}{3} \sum_i r_{t,i}^4 \rightarrow IQ_t$, the scale of the measurement error of RV
 - ▶ The limit needs a continuous, noise-free price and an equally spaced grid; a jump ΔJ adds about $M(\Delta J)^4/3$, so raw RQ is jump-sensitive
- Bipower variation (robust to jumps), realised kernel (robust to noise) [Barndorff-Nielsen et al. \(2008\)](#), range-based estimators
- Target of this chapter: five-minute RV (rv5)
 - ▶ Across 400 assets, little beats five-minute RV as a proxy of daily variance [Liu, Patton & Sheppard \(2015\)](#)
 - ▶ Its semivariances and quarticity are on the same grid, so SHAR and HARQ use consistent inputs

Targets and Losses

- Horizon h : $RV_{t+1:t+h} = \frac{1}{h} \sum_{j=1}^h RV_{t+j}$, $h \in \{1, 5, 22\}$; log models forecast $\ln RV_{t+1:t+h}$
 - ▶ Back to levels with $F_t = \exp(\hat{y}_t + \hat{\sigma}^2/2)$, $\hat{\sigma}^2$ = residual variance: exact if the log error is conditionally Normal with constant variance, otherwise an approximation
 - ▶ For $h > 1$ targets overlap: this induces serial dependence (through lag $h - 1$ by construction, possibly longer), so every test uses a HAC variance
- A loss is **robust** if, for a conditionally unbiased proxy ($\mathbb{E}_{t-1}[RV_t] = \sigma_t^2$), the *expected*-loss ranking with RV_t equals that with the true variance [Patton \(2011\)](#)
 - ▶ $MSE_t = (RV_t - F_t)^2$, $QLIKE_t = RV_t/F_t - \ln(RV_t/F_t) - 1 \geq 0$ (zero only if $F_t = RV_t$)
- QLIKE depends on the ratio RV_t/F_t : the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days; QLIKE is the primary loss, MSE the secondary one

VOLARE: an Open Archive of Realised Measures

- Built at the Universities of Messina and Florence from tick data [Cipollini et al. \(2026\)](#)
 - ▶ Asset-specific cleaning, trading calendars and timestamp precision; the same estimators for every asset class
- Free for research at volare.unime.it; not redistributed with this course
 - ▶ Download the realised-variance files and point the notebooks to your local folder
- Citation requested by the authors: Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732



University of Messina, the home of VOLARE

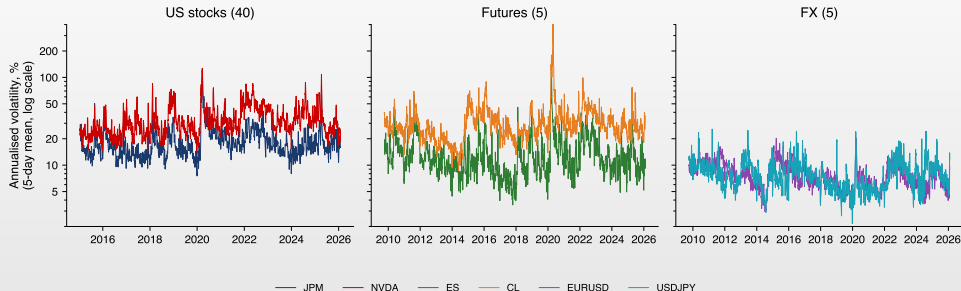
Photo: [Superchilum \(2013\)](#); CC BY-SA 3.0; [Wikimedia Commons](#)

The Panel: 50 Assets in Three Classes

Class	Assets	Sample	Observations	Median volatility
American large-cap stocks	40	2 Jan 2015 – 30 Jan 2026	111 440	17.1%
Futures (C, CL, ES, GC, NG)	5	28 Sep 2009 – 30 Jan 2026	21 059	19.7%
FX (AUD, EUR, GBP, CAD, JPY vs USD)	5	25 Sep 2009 – 30 Jan 2026	21 203	7.3%

- Stocks: 40 members of the Dow Jones 30 and S&P 100 (AAPL to XOM); futures: corn, crude oil, E-mini S&P 500, gold, natural gas
- Median volatility: $\sqrt{252 \text{ median}(RV_t)}$ per asset, then the median over the class; weekends are absent in all three files
- Variables used: `rv5`, `rsp5`, `rsn5`, `rq5`, and the daily open and close prices

Realised Volatility Across Asset Classes



- Annualised volatility $\sqrt{252 RV_t}$, five-day mean, log scale, two assets per class; typical levels: FX about 7.3%, stocks about 17.1% a year
- All classes share March 2020; crude oil (CL) peaks in April 2020, when the front contract traded below zero: a forecaster must adapt within days
- On a log scale volatility is stationary and persistent (the long memory HAR approximates); spikes decay within weeks, so the shape of the recent path carries information

HAR: Three Averages of the Past

- HAR (Heterogeneous AutoRegressive) model [Corsi \(2009\)](#)

- ▶ $RV_{t+1:t+h} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$
- ▶ $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$, $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

- ▶ $\phi_1 = \beta_d + \beta_w/5 + \beta_m/22$, $\phi_{2..5} = \beta_w/5 + \beta_m/22$, $\phi_{6..22} = \beta_m/22$

- Traders at daily, weekly and monthly frequencies: a cascade that mimics long memory with three parameters

- log-HAR: $\ln RV_{t+1:t+h}$ on $\ln RV_t$, $\ln RV_t^{(w)}$, $\ln RV_t^{(m)}$; no negative forecasts, less driven by spikes

- ▶ Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in $\ln RV$ (Seminar 20, A5)

Two Refinements: HARQ and SHAR

- RV is a generated regressor: $RV_t = IV_t + \eta_t$, $\text{Var}(\eta_t) \approx 2IQ_t/M$; errors-in-variables attenuate β_d on noisy days
 - ▶ HARQ [Bollerslev, Patton & Quaedvlieg \(2016\)](#):
 $RV_{t+1:t+h} = \beta_0 + (\beta_d + \beta_Q(RQ_t^{1/2} - \overline{RQ^{1/2}}))RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$, expected $\beta_Q < 0$
 - ▶ BPQ filter: a forecast outside the in-sample range of the target is replaced by the in-sample mean (“insanity filter”)
- SHAR [Patton & Sheppard \(2015\)](#): split the daily term by the sign of the intraday returns
 - ▶ $RV_{t+1:t+h} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$
 - ▶ Patton and Sheppard find $\beta^- > \beta^+$: an intraday version of the leverage effect (Chapter 5)
- The signature can capture a related effect from daily data: the Lévy area between returns and log RV (Path Signatures section)

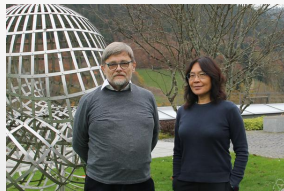
Seven Models in One Table

Model	Target	Weights	Regressors
HAR	RV	equal	$RV_t, RV_t^{(w)}, RV_t^{(m)}$
log-HAR	$\ln RV$	equal	the same, in logs
HARQ	RV	equal	$HAR + RV_t (RQ_t^{1/2} - \overline{RQ^{1/2}})$
SHAR	RV	equal	$RV_t^+, RV_t^-, RV_t^{(w)}, RV_t^{(m)}$
log-HAR-K	$\ln RV$	signature kernel	log-HAR regressors
Sig-L	$\ln RV$	equal	log-HAR + 39 signature terms, two-step LASSO
Sig-LK	$\ln RV$	signature kernel	as Sig-L: the method of Gu et al.

- The last four form a 2×2 ablation: signature features (yes/no) $\times$ kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models: same target, and its regressors are among the candidates; but the LASSO may also drop log-HAR terms, so Sig-L vs log-HAR compares two procedures, not only two feature sets

Where Signatures Come From

- **Chen (1957)**: iterated integrals of a path form an algebra and determine it up to reparametrisation
 - ▶ Kuo-Tsai Chen, geometry of loop spaces
- **Lyons (1998)**: rough path theory; the signature is the natural feature set for controlled differential equations
 - ▶ A response to a path depends on it (to first order) only through its signature
- **Machine learning**: [Levin, Lyons & Ni \(2013\)](#), [Chevyrev & Kormilitzin \(2016\)](#), [Lyons & McLeod \(2022\)](#); finance: signature-based asset price models [Cuchiero, Gazzani & Svaluto-Ferro \(2023\)](#)



Terry Lyons (left), Oberwolfach, 2024

Photo: [Katrin Schmid \(2024\)](#); CC BY-SA 2.0 DE; [Wikimedia Commons](#)

A Window of Data Is a Path

- Take the last L days of $d - 1$ series and connect the points linearly
 - ▶ $X : [0, 1] \rightarrow \mathbb{R}^d$, $X_{u_i} = (u_i, \ln \text{RV}_{t-L+1+i}, \sum_{j \leq t-L+1+i} r_j)$, $u_i = i/(L - 1)$
- **Time augmentation:** the first channel is time itself
 - ▶ Without it the signature cannot tell a fast move from a slow one (reparametrisation invariance)
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
 - ▶ Example: RV rose while prices fell, or RV rose first and prices fell later

The Signature: Definition

- For a path $X : [a, b] \rightarrow \mathbb{R}^d$ of bounded variation and a word $(i_1, \dots, i_k)$ with letters in $\{1, \dots, d\}$
 - ▶ $S(X)_{a,b}^{i_1 \dots i_k} = \int_{a < u_1 < \dots < u_k < b} dX_{u_1}^{i_1} \dots dX_{u_k}^{i_k}$ (iterated integral)
- The signature is the whole sequence of levels
 - ▶ $\text{Sig}(X) = (1, S^{(1)}, S^{(2)}, S^{(3)}, \dots)$, $S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$ has d^k terms
- Level 1: $S^i = X_b^i - X_a^i$, the total increments
 - ▶ Level 2: $S^{ij} = \int_a^b (X_u^i - X_a^i) dX_u^j$; level 3 adds one more integral

Linear Segments and Chen's Identity

- A linear segment with increment $\Delta \in \mathbb{R}^d$ has the tensor exponential as signature
 - ▶ $\text{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = (1, \Delta, \frac{1}{2!}\Delta^{\otimes 2}, \frac{1}{3!}\Delta^{\otimes 3}, \dots)$
- **Chen's identity** [Chen \(1957\)](#): the signature of a concatenation is the tensor product
 - ▶ $\text{Sig}(X * Y) = \text{Sig}(X) \otimes \text{Sig}(Y), \quad (A \otimes B)^{(k)} = \sum_{p=0}^k A^{(k-p)} \otimes B^{(p)}$
- A piecewise-linear path with m segments: $\text{Sig} = \exp_{\otimes}(\Delta_1) \otimes \dots \otimes \exp_{\otimes}(\Delta_m)$
 - ▶ This is the whole algorithm: m updates, each costing $O(d^N)$ at depth N (a few lines of numpy)
 - ▶ Rolling windows: Chen also gives $\text{Sig}(X_{[s,t]}) = \text{Sig}(X_{[a,s]})^{-1} \otimes \text{Sig}(X_{[a,t]})$

Worked Example: a Two-Segment Path

Path

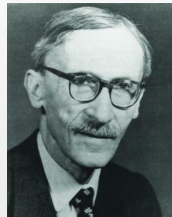
- ▣ $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2); \Delta_1 = (1, 0), \Delta_2 = (0, 2)$
- ▣ Level 1: $S^{(1)} = \Delta_1 + \Delta_2 = (1, 2)$
- ▣ Level 2 by Chen: $S^{ij} = \frac{1}{2}\Delta_1^i\Delta_1^j + \Delta_1^i\Delta_2^j + \frac{1}{2}\Delta_2^i\Delta_2^j$

Result

- ▣ $S^{11} = \frac{1}{2}, S^{12} = 0 + 1 \cdot 2 + 0 = 2$
- ▣ $S^{21} = 0, S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- ▣ Check: $S^{11} = (\Delta X^1)^2/2$ and $S^{22} = (\Delta X^2)^2/2$ (shuffle identity)
- ▣ $S^{12} \neq S^{21}$: the order matters (first right, then up)

The Lévy Area: Order as a Number

- Antisymmetric part of level 2
 - ▶ $A^{ij} = \frac{1}{2} (S^{ij} - S^{ji}) =$ signed area between the path and its chord
- Symmetric part: $\frac{1}{2} (S^{ij} + S^{ji}) = \frac{1}{2} \Delta X^i \Delta X^j$, no new information
 - ▶ Example above: $A^{12} = (2 - 0)/2 = 1$, the area of the triangle under the path
- Channels (return, log RV): a price fall followed by a volatility rise gives $A < 0$, but so does a rise followed by a volatility fall
 - ▶ A lead-lag statistic: leverage-type sequences (Chapter 5) are one contributor to a negative area; the sign alone does not identify them

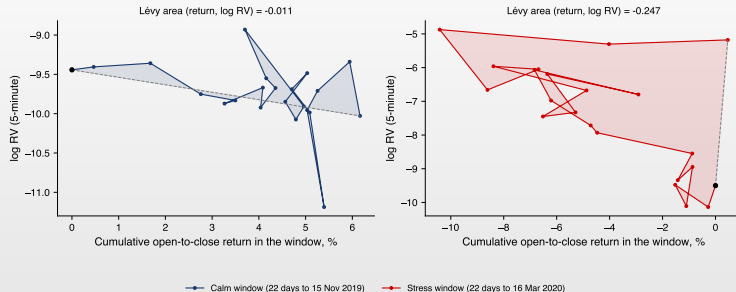


Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE;

[Wikimedia Commons](#)

Two Windows of JPM as Paths

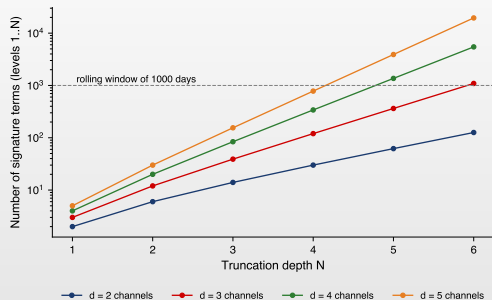


- Path (cumulative open-to-close return, log RV) over 22 days; dashed chord from start to end; the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019): log RV moves by -0.58, the path wanders, Lévy area -0.011; stress window (to 16 Mar 2020): log RV rises by +4.32 ($RV \times 75$) while returns fall, Lévy area -0.247
- The net intraday return is only +0.5%: the crash was largely overnight, outside this channel; no signature level recovers what the input omits

Why Signatures Work: Three Theorems

- **Uniqueness** [Hamblly & Lyons \(2010\)](#): the full $\text{Sig}(X)$ determines X up to tree-like equivalence and translation; with a strictly increasing time channel, up to translation only
 - ▶ A finite truncation Sig^N is not injective
- **Shuffle identity**: products of signature terms are linear combinations of higher terms, $S^i S^j = S^{ij} + S^{ji}$,
 $S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$
 - ▶ So a linear model in $\text{Sig}^N(X)$ already contains polynomials of the increments
- **Universality**: for a compact set K (1-variation topology) of time-augmented paths with a common start and $f : K \rightarrow \mathbb{R}$ continuous, for every $\varepsilon > 0$ there is a linear ℓ with $\sup_{X \in K} |f(X) - \langle \ell, \text{Sig}(X) \rangle| < \varepsilon$ (Stone–Weierstrass, via the shuffle identity)
 - ▶ This is why a LASSO on signatures can approximate nonlinear functions of the path; the price is d^N features, so sparsity is needed
- **Factorial decay**: $\|S^{(k)}\| \leq \|X\|_{\mathbf{1}\text{-var}}^k / k!$: truncating at a small depth N loses little for paths of moderate length

How Many Features? The Truncation Dimension



- Levels 1..N of a d -dimensional path:

$$M(d, N) = \sum_{k=1}^N d^k = \frac{d^{N+1} - d}{d - 1} \text{ terms}$$
- This chapter: $d = 3$ channels (time, log RV, cumulative return), depth $N = 3$: $3 + 9 + 27 = 39$ terms
 - ▶ Three are constant (words with only the time letter), so 36 vary
- $d = 5$, $N = 4$ would give 780 terms for 1000 observations: the LASSO would select from noise
- Gu et al. do not disclose N and L ; here $N = 3$, $L = 22$ are fixed ex ante, with $N = 2$ as a robustness check

Two Invariances to Keep in Mind

- ▣ **Translation invariance:** $\text{Sig}(X + c) = \text{Sig}(X)$ — only increments enter
 - ▶ The signature does not see the **level** of volatility, only the shape of its recent path
 - ▶ So the level must enter separately: the log-HAR terms play the role of the external factors x_T of Gu et al.
- ▣ **Reparametrisation invariance:** running the path faster or slower leaves Sig unchanged
 - ▶ Removed by the time channel; basepoint augmentation (prepending a segment from 0 to the original starting point) would remove the first invariance [Morrill et al. \(2020\)](#)
 - ▶ Translating the path to start at 0 does not: it is itself a translation (Seminar 20, C1 uses the prepended segment)

The Signature Kernel

- Kernel = inner product of feature maps [Király & Oberhauser \(2019\)](#)
 - ▶ $k_N(X, Y) = \langle \text{Sig}^N(X), \text{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^N \sum_{|w|=k} S(X)^w S(Y)^w$
- Distance induced by the kernel (Gu et al., Eq. 9)
 - ▶ $d_{\text{Sig}}(X, Y) = k(X, X) - 2k(X, Y) + k(Y, Y) = \|\text{Sig}^N(X) - \text{Sig}^N(Y)\|^2$
- Gu et al. and this chapter use this **truncated** kernel, with N as for the features
 - ▶ The untruncated kernel solves a Goursat PDE (partial differential equation) [Salvi et al. \(2021\)](#); the paper cites it but does not use it
- Channels are rescaled before the kernel (Appendix): otherwise the channel with the largest units dominates

The Problem at Amazon and the Features

- Weekly North American truckload rates, national and five regions, 2018–2022, forecast up to 12 weeks ahead [Gu et al. \(2024\)](#)
 - ▶ More than 100 candidate factors screened to about 40; in Amazon operations since November 2022; KDD 2024 (ACM Conference on Knowledge Discovery and Data Mining)
- Features at week τ (Section 4.1, Eq. 14): external factors x_τ and the signature of the time-augmented target path
 - ▶ $\xi_\tau = [x_\tau, \text{Sig}^N(\hat{y}_{\tau-l:\tau})]$, $\hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$
- One direct regression per horizon, $y_{\tau+\Delta t}$ on ξ_τ ; lookback l , depth N , temperature γ are not published (confidentiality)



A container truck on an American interstate, 2025

Photo: [DanTD \(2025\)](#); CC BY 4.0; [Wikimedia Commons](#)

The Method: Two-Step LASSO

- ▣ Selection by a weighted LASSO (Least Absolute Shrinkage and Selection Operator) [Tibshirani \(1996\)](#) (Eq. 14)
 - ▶ $\hat{\theta}_L = \arg \min_{\theta} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2 + \lambda \|\theta\|_1$
- ▣ Refit by weighted OLS on the selected support (Eq. 15)
 - ▶ $\hat{\theta} = \arg \min_{\text{supp}(\theta)=\text{supp}(\hat{\theta}_L)} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2$
- ▣ Why refit: LASSO shrinks the kept coefficients towards zero; OLS after selection removes that bias [Belloni & Chernozhukov \(2013\)](#)
 - ▶ The paper calls it “adaptive” because of the weights w_{τ} , not in the sense of the adaptive LASSO of [Zou \(2006\)](#)
- ▣ How λ is chosen is not reported

The Method: Kernel Weights

- ▣ Kernel path: the *pairs* $z_\tau = (x_\tau, y_{\tau+\Delta t})$ of predictors and later target, not the features ξ_τ (Algorithm 1)
 - ▶ Distance to the last window with an observed target (Eq. 11): $\delta_\tau = d_{\text{Sig}, N}(z_{\tau-l:\tau}, z_{t-\Delta t-l:t-\Delta t})$
- ▣ Softmax weights with temperature $\gamma \geq 0$ (Eq. 12)
 - ▶ $w_\tau = \frac{\exp(-\gamma \delta_\tau)}{\sum_s \exp(-\gamma \delta_s)}$
 - ▶ $\gamma = 0$: ordinary (equal-weight) regression; $\gamma \rightarrow \infty$: only the most similar past window
- ▣ A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
 - ▶ Cost of the idea: fewer effective observations, $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$ (Kish)

What the Paper Reports, Read Critically

- National rates, three months ahead, April–November 2021 (Table 1): relative error $|y - \hat{y}|/|y|$ of about 1–2% vs 13–27% for the industry forecast, “more than fivefold”
 - ▶ Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
 - ▶ Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it
- Limits of the evidence
 - ▶ **One benchmark:** an unnamed commercial forecast; no ARIMA, no LASSO without signatures, no random walk
 - ▶ **Short evaluation:** a few months, a few series, no standard errors, no test of equal accuracy
 - ▶ **Undisclosed tuning:** l , N , γ , λ and how they were chosen
- None of this makes the method wrong; it makes the evidence thin, which is why it is tested elsewhere, replication before extension (Chapter 19)

What Transfers to Volatility, and What Does Not

Transfers

- ▣ Signature of the time-augmented recent path of the target
- ▣ Two-step LASSO with weights, direct regression per horizon
- ▣ Softmax weights $\exp(-\gamma \delta_\tau)$ with the truncated signature kernel

Must be adapted

- ▣ Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- ▣ External factors x_τ replaced by the log-HAR terms (they carry the level)
- ▣ Undisclosed choices (l, N, γ, λ) fixed ex ante in this chapter
- ▣ Kernel: the path of this chapter (time, $\ln RV$, return) ending at τ vs the window ending at t , updated every 5 days, instead of the pairs $(x_\tau, y_{\tau+\Delta t})$ ending at $t - \Delta t$: an adaptation
- ▣ Strong benchmarks and formal tests instead of one commercial forecast

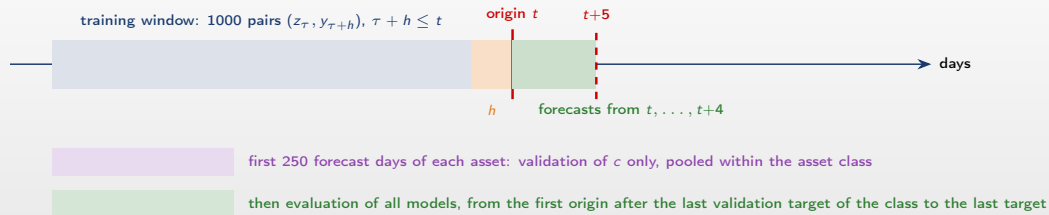
Hypotheses

- ▣ **H1 (features):** signature terms add information beyond log-HAR
 - ▶ Test: Sig-L vs log-HAR, same target, same equal weights
- ▣ **H2 (weights):** kernel weights improve a given regression
 - ▶ Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- ▣ **H3 (method):** the full method of Gu et al. beats the HAR family
 - ▶ Tests: Sig-LK vs HAR and vs log-HAR; model confidence set of all seven models
- ▣ **H4 (stress):** any gain is larger in stress periods (COVID-19 crash, April 2025), where similar past windows should matter most

Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature: $\gamma = c / \text{median}_{\tau}(\delta_{\tau})$ in every window
 - ▶ $c = 1$: a window at the median distance gets weight e^{-1} relative to an identical one
- Grid $c \in \{0.5, 1, 2, 4\}$, chosen per horizon and asset class on the first 250 forecast days of each asset, pooled within the class
 - ▶ Criterion: mean over the class of $\text{QLIKE}(c)$ divided by the asset's average over the grid
 - ▶ Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the flattest value, $c = 0.5$, wins at every horizon for Sig-LK; for futures and FX the choice changes with the horizon: a flat, noisy criterion (all values in the Appendix)

Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs; between re-estimations the coefficients are fixed but the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin t

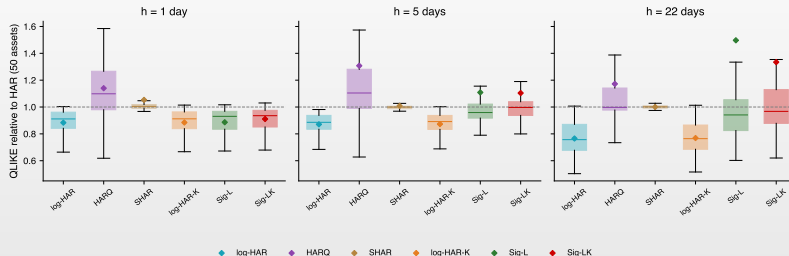
Evaluation Protocol

- Per asset and horizon: mean QLIKE and MSE; DM (Diebold–Mariano) test with Newey–West variance, bandwidth $\max(h, \lceil T^{1/3} \rceil)$ [Diebold & Mariano \(1995\)](#)
 - ▶ Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test; the conditional GW test uses instruments $(1, d_{t-h})$ [Giacomini & White \(2006\)](#)
- MCS (model confidence set) at 90%, $T_{\max}$ statistic, stationary bootstrap, 1000 replications [Hansen, Lunde & Nason \(2011\)](#)
 - ▶ Block length $\max(h, 10)$ days
- Panel: DM test on the cross-sectional mean loss differential (one series per date)
 - ▶ Counting wins over 50 assets: Holm (FWER) and BH (FDR) corrections [Benjamini & Hochberg \(1995\)](#)
- Subperiods: 20 Feb–30 Apr 2020, 2–30 Apr 2025, all other days; the paper's relative error as a descriptive statistic
- 265 929 out-of-sample forecasts per model; stocks are evaluated from January 2020 (the COVID-19 crash is in every stock sample), futures and FX from 2014 (dates in the Appendix)

A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
 - ▶ A pipeline test on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
 - ▶ On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before the panel run: primary results with positivity only (a forecast ≤ 0 becomes the in-sample mean)
 - ▶ The BPQ filter is kept as a robustness check and reported in full
- Rule: a deviation is allowed, a hidden deviation is not; state what was seen when it was changed

QLIKE Relative to HAR, Asset by Asset



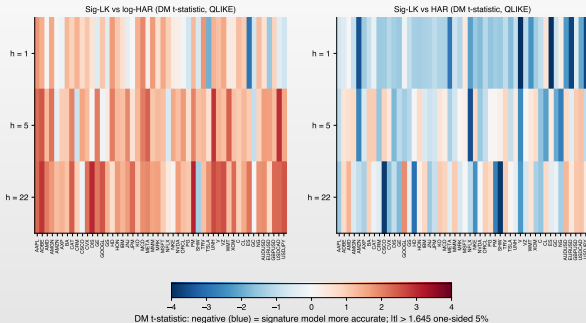
- Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges
- $h = 1$: log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886); Sig-LK is +4.5% relative to log-HAR: no gain from 36 extra features
- $h = 5, 22$: Sig-LK falls behind log-HAR (+28.0%, +92.9%); its medians (0.997, 0.968) are much better than its means: a few assets and days drive the loss
- HARQ and SHAR lose to HAR in levels here (Appendix: the BPQ filter); first verdict: the method does not beat the simplest log model (a near-tie is not a proof of equal accuracy)

Asset by Asset: Wins and Losses, Corrected

Comparison (QLIKE)	$h = 1$		$h = 5$		$h = 22$	
Sig-LK vs log-HAR	2/0/0	5/0/0	0/0/0	15/0/2	0/0/0	27/1/16
Sig-L vs log-HAR	3/1/1	5/0/0	0/0/0	9/0/0	1/0/0	21/0/0
log-HAR-K vs log-HAR	3/0/0	3/0/0	3/0/0	1/0/0	5/0/0	6/0/0
Sig-LK vs Sig-L	2/0/0	5/0/0	0/0/0	5/0/0	2/0/0	10/0/0
Sig-LK vs HAR	12/5/6	0/0/0	7/1/1	0/0/0	8/2/4	1/0/0
log-HAR vs HAR	21/5/17	0/0/0	23/13/20	0/0/0	42/25/42	0/0/0

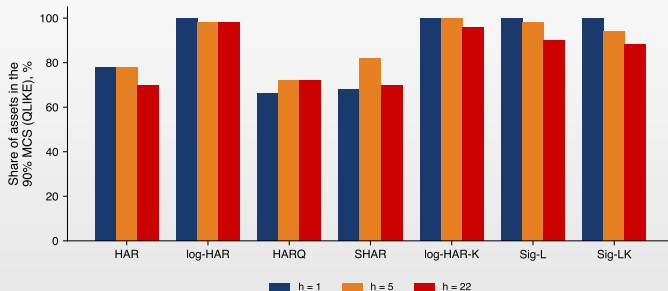
- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets; one-sided DM at 5%, family of 50 assets; Holm controls the FWER under any dependence, BH the FDR under independence or PRDS [Benjamini & Yekutieli \(2001\)](#)
- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at $h = 1$ raw, 1 after Holm; about 2.5 raw wins are expected by chance
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at $h = 1$, Sig-LK wins on 12 assets, but log-HAR alone wins on 21: choose the benchmark that isolates the idea

DM Statistics for Every Asset and Horizon



- Diebold–Mariano t -statistics of the QLIKE differential; blue: Sig-LK more accurate, red: less accurate
- Against log-HAR (left): pale at $h = 1$ (blue exceptions ES, TSLA, EURUSD; five assets significantly red before correction), red at $h = 5, 22$
- Against HAR (right): blue at $h = 1$ because Sig-LK inherits the log target; one cell such as ES at $h = 1$ would make signatures look like a clear success

Model Confidence Sets



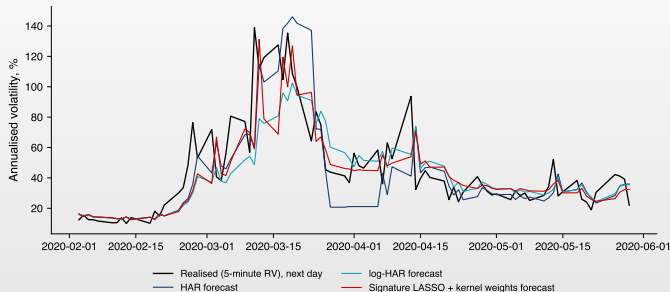
- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE: log-HAR 100%, 98%, 98%; Sig-LK 100%, 94%, 88% ($h = 1, 5, 22$)
- The MCS rarely separates the four log models; HAR, HARQ and SHAR are excluded for a fifth to a third of the assets: the level target is the clearest loser
- Under MSE almost every model stays in the set [Patton \(2011\)](#): small differences in average loss need very long samples [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)

Panel Tests: Pooling the Evidence

Comparison: DM t (one-sided p)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	+1.45 (0.927)	+1.68 (0.953)	+1.80 (0.964)
Sig-L vs log-HAR	-1.91 (0.028)	+1.48 (0.931)	+1.51 (0.935)
log-HAR-K vs log-HAR	+1.89 (0.971)	+0.40 (0.654)	+1.92 (0.973)
Sig-LK vs Sig-L	+1.74 (0.959)	+0.98 (0.837)	-0.77 (0.221)
log-HAR vs HAR	-3.09 (<0.001)	-3.95 (<0.001)	-4.50 (<0.001)

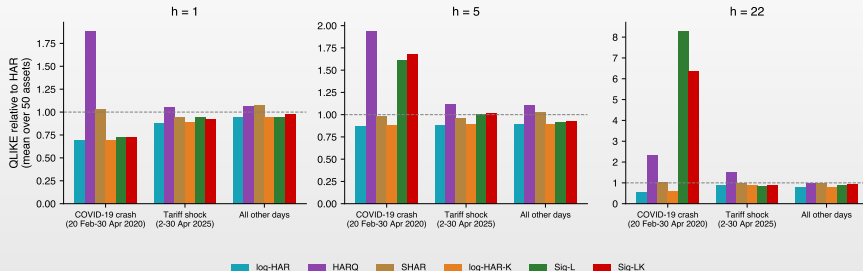
- ▣ Series: the mean QLIKE differential over all assets trading that day; negative t = first model better
- ▣ Pooled, only the log transform is clearly significant; Sig-L vs log-HAR at $h = 1$ is borderline ($p = 0.028$), driven by futures
 - For the 40 stocks alone the sign flips: $t = +1.72$ against Sig-L
- ▣ Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

Inside the COVID-19 Crash: JPM, Next Day



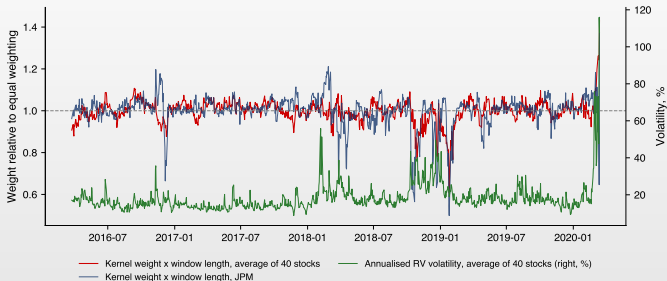
- Annualised volatility $\sqrt{252F_t}$ of the forecasts and $\sqrt{252RV_{t+1}}$ of the target, February–May 2020; all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March, when the path has already turned; after the peak both decay at similar speeds
- HAR in levels overshoots, then turns negative in late March: the positivity rule replaces it by the in-sample mean (the flat segment)

Stress Periods vs Normal Times (H4)



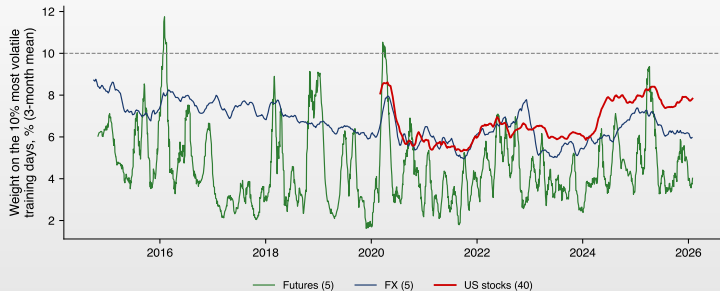
- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$, COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89; April 2025: 0.88 and 0.92; other days: 0.94 and 0.97
- $h = 22$, COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR: exactly where similar past windows should help, the signature models fail most; H4 not supported (descriptive, no formal interaction test)

Kernel Weights at the Origin 16 March 2020



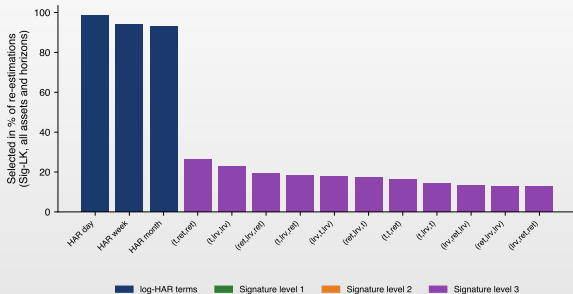
- Weights of Sig-LK ($h = 1$) at the origin 16 March 2020 over the training days (24 Mar 2016 – 13 Mar 2020), times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times $1/n$, effective sample size 975 of 1000; the highest weights go to the last days before the origin
- The late-2018 sell-off gets *less* weight; weights are almost uncorrelated with log volatility (+0.04): the translation-invariant signature sees shape, not level

Does the Kernel Ever Favour Stress?



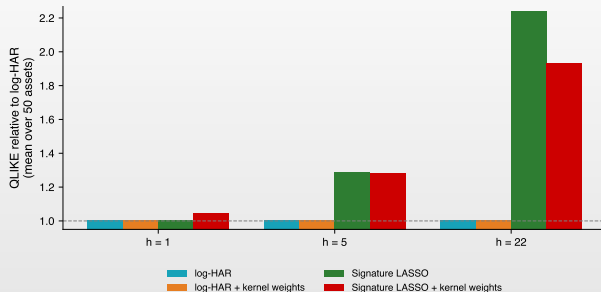
- Weight that Sig-LK ($h = 1$) puts on the 10% most volatile days of its training window, averaged by class, three-month mean
- Stocks: mean 6.7%, between 4.4% and 9.9% (maximum on 17 Mar 2020): always below the 10% of equal weighting; futures and FX show the same pattern
- The mechanism behind the null result for H2: the weights do not implement “learn from past crises”

What the LASSO Selects



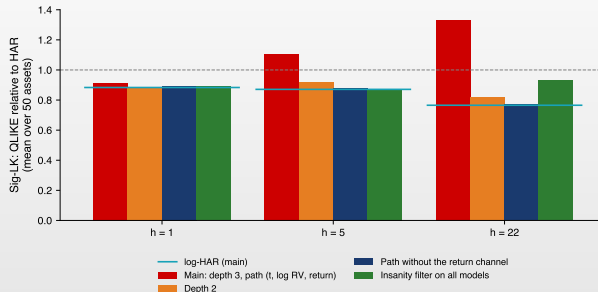
- Share of re-estimations in which a regressor is in the BIC-selected support of Sig-LK; t = time, lrv = $\ln RV$, ret = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations: the LASSO rediscovers HAR; average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term, (t,r,r) (27%), and all frequent ones are of level 3: cubic in the increments, far outside the training range in a crash, amplified by $\exp(\cdot)$

The 2×2 Ablation



- Mean over the 50 assets of QLIKE relative to log-HAR: features (signature or not) crossed with weights (kernel or equal)
- Weights alone (log-HAR-K): 1.002, 1.002, 1.004, no effect (average effective sample size 816 of 1000); features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK: 1.045, 1.280, 1.929) damp the one-month failure without removing it and worsen $h = 1$ through crude oil (CL: 2.71); Gu et al. (Table 3) saw a kernel gain in six weekly values of one region

Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets) in the main design and in three pre-registered variants; the line marks log-HAR
- Depth 2: 0.922 and 0.820 at $h = 5, 22$ (main: 1.104, 1.334); path without returns: 0.878 and 0.770: the long-horizon failures come from level-3 terms with the return channel
- The stable variants only match log-HAR (0.871, 0.765): the null result for H1 does not depend on depth or channels, the instability does

By Asset Class

Sig-L relative to log-HAR (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Stocks (40)	1.009	1.332	2.505
Futures (5)	0.965	1.084	1.206
FX (5)	0.996	1.111	1.134

- The only gain: futures at $h = 1$ (-3.5%) and, marginally, FX
 - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

Limitations and Open Directions

Limitations of this study

- One design out of many ($L = 22$, $N = 3$, three channels, BIC, a four-point grid for c): a null result for this design is not one for signatures in general
- Truncated kernel only; no external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#) and machine-learning benchmarks (Chapter 13) are outside the comparison; stocks are evaluated over six years only

Open directions

- A level-aware kernel: basepoint augmentation or the level of $\ln RV$ in the distance (Seminar 20, C1)
- Signatures of the intraday path (5-minute returns); the PDE kernel [Salvi et al. \(2021\)](#); log-signatures and bounded (tanh) channels against explosive level-3 terms [Morrill et al. \(2020\)](#)
- Cross-asset paths, where Lévy areas measure lead-lag across markets; futures at $h = 1$ on a longer panel, with this design fixed in advance

An Open Question

When does a method transfer across domains?

- Methods travel faster than evidence: a KDD paper on freight rates becomes a candidate for volatility, credit, energy
 - ▶ Which properties of the source problem made the method work, and does the target share them?
- Finance adds a hard filter: 164 teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#)
- Formal version: a transfer succeeds if the gain over the strongest simple benchmark survives a pre-registered set $\mathcal{S}$ of designs, assets and horizons, with the family corrected
 - ▶ In this chapter: $\mathcal{S} = 50 \text{ assets} \times 3 \text{ horizons} \times 4 \text{ variants}$; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells; the gain did not survive

The Discovery Loop Applied to This Chapter

- **Read:** an assistant lists every equation, hyperparameter and benchmark of the paper and marks what is not disclosed; each item is checked against the PDF
 - ▶ **Map:** for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register:** hypotheses, grid, tests and correction committed before the first result; the assistant drafts, the team decides every number
- **Build and test the pieces:** signatures against identities (level 1 = increment, Chen on a split path), not against another library
- **Run the grid, then attack it:** which conclusions survive Holm over 50 assets, which single asset would make a misleading headline?
- **Explain the null:** here flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data; the full prompts and the researcher's checklist are in the Appendix

Mini-Case: Three Headlines From the Same Runs

- “Signatures beat HAR on 12 of 50 assets” (one-day horizon, raw DM)
 - ▶ True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- “Signatures improve E-mini S&P 500 volatility forecasts” (ES, $h = 1$, DM $t = -2.91$ against log-HAR)
 - ▶ One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at $h = 1$ and none at $h = 5, 22$
- The headline of the pre-registered evaluation: “no gain over log-HAR; worse at longer horizons; kernel weights inert”
 - ▶ An AI assistant asked to “find where the method works” will produce the first two; the plan protects the third

Your Turn

- Pick one open direction of this chapter (level-aware kernel, intraday signatures, log-signatures, futures only)
 - ▶ Write H_0/H_1 , the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Commit the pre-registration before the first out-of-sample number; run the grid; report the distribution over $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C2 gives typical ones)

Key Takeaways

- Signatures turn a window into a feature vector with a theory behind it: Chen's identity, uniqueness, universality
- Transferred to realised variance, the method of Gu et al. does not beat log-HAR
 - ▶ H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
 - ▶ H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- The translation-invariant kernel sees shape, not level; level-3 terms explode in crashes
- The strongest effect in the study is the oldest one: model the log of RV
- A pre-registered null result, with ablations and a mechanism, is a finding

Further Study

Reading

- ▣ Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022); realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python
- ▣  Machine Learning in Financial Risk

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: VOLARE, free for research at volare.unime.it

Appendix: Features and Scaling

- Regressors at day τ for the signature models: 3 log-HAR terms + 39 signature terms
 - ▶ $\xi_\tau = [\ln \text{RV}_\tau, \ln \text{RV}_\tau^{(w)}, \ln \text{RV}_\tau^{(m)}, \text{Sig}^3(X_{\tau-21:\tau})]$, path $X = (\text{time}, \ln \text{RV}, \text{cumulative } r)$, 22 points
- Non-time channel c divided by $s_c \sqrt{L-1}$, $s_c =$ standard deviation of its daily increments in the training window up to the origin t
 - ▶ Time is not rescaled (constant increments, $s = 0$): it runs from 0 to 1; a word $(i_1..i_k)$ is rescaled by $\prod_j c_{ij}$, so signatures are computed once
- Before the LASSO every column is standardised with weighted moments of the training window; constant columns (words of time only) are dropped
- Raw features at τ use data up to τ ; scales and standardisation use data up to the origin t : the no-look-ahead cutoff is t

Appendix: Estimation: Choosing λ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS (least-angle regression)
 - ▶ For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion (Bayesian information criterion) with the Kish effective sample size, a heuristic of this design
 - ▶ $\text{BIC}(S) = n_{\text{eff}} \ln \hat{S}_S^2 + (|S| + 1) \ln n_{\text{eff}}, \quad n_{\text{eff}} = 1 / \sum_{\tau} w_{\tau}^2$
- Why not cross-validation: 50 assets $\times$ 3 horizons $\times$ hundreds of re-estimations; BIC uses only the window
 - ▶ Kish size corrects for unequal weights, not for serial dependence (overlapping targets, $h > 1$): not a likelihood-based BIC for these data

Appendix: Kernel Temperatures Chosen on the Validation Block

- Chosen c for Sig-LK ($h = 1, 5, 22$)
 - ▶ stocks 0.5, 0.5, 0.5; futures 2.0, 1.0, 0.5; FX 0.5, 2.0, 0.5
- Chosen c for log-HAR-K ($h = 1, 5, 22$)
 - ▶ stocks 0.5, 0.5, 1.0; futures 2.0, 1.0, 4.0; FX 4.0, 2.0, 4.0
- Grid $c \in \{0.5, 1, 2, 4\}$; the flattest value wins for the stocks at every horizon; for futures and FX the choice changes with the horizon

Appendix: What Was Evaluated

Class	Evaluation starts	Ends	Days per asset (median)
Stocks	27 Jan 2020	29 Jan 2026	1 499
Futures	25 Sep 2014	29 Jan 2026	2 919
FX	27 Aug 2014	29 Jan 2026	2 952

- 265 929 out-of-sample forecasts per model ($50 \text{ assets} \times 3 \text{ horizons}$)
- Stocks start in 2015, so their evaluation begins in January 2020: the COVID-19 crash is in every stock sample
 - Futures and FX start in 2009 and are evaluated from 2014
- Computation: 18.5 core-hours; 87 minutes on 14 parallel workers; signatures and LASSO in numpy and scikit-learn

Appendix: The Paper's Metric on VOLARE

Median relative error $ RV - F /RV$ (%)	$h = 1$	$h = 5$	$h = 22$
HAR	57.2	48.9	51.7
log-HAR	50.4	41.9	38.7
HARQ	55.0	49.0	50.7
SHAR	57.9	49.2	51.8
log-HAR-K	49.7	41.0	38.7
Sig-L	50.1	42.6	42.6
Sig-LK	50.6	43.6	43.7

- Median over assets of the mean relative error: about 40–60%, against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target: the evaluated models have much larger relative errors than those reported for freight rates, without establishing a lower bound on attainable error
 - ▶ Sig-LK forecasts above ten times the in-sample maximum: 8, 29, 36 ($h = 1, 5, 22$), on 10 assets at $h = 5$; log-HAR: none

Appendix: The Insanity Filter, Revisited

QLIKE relative to HAR, BPQ filter on all models	$h = 1$	$h = 5$	$h = 22$
HAR	1.000	1.000	1.000
log-HAR	0.763	0.654	0.778
HARQ	0.995	0.912	1.014
SHAR	1.014	1.003	1.000
log-HAR-K	0.772	0.659	0.779
Sig-L	0.871	0.819	0.904
Sig-LK	0.890	0.867	0.930

- The filter rescues HARQ relative to HAR (0.995 at $h = 1$ instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209 (unfiltered 1.045, 1.280, 1.929): the gap widens at $h = 1, 5$ and narrows at $h = 22$
- Filtered forecasts, $h = 1$: HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893): the filter hits the signature models far more often than log-HAR

Appendix: What the Researcher Must Check

- **The method is the paper's:** kernel, weights, refit, as in the cited equations
 - ▶ Test: every design slide cites an equation, table or section of the paper; undisclosed choices are labelled as choices of the study
- **No look-ahead:** λ , scales, c and features use only data available at the origin
 - ▶ Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change
- **The benchmark isolates the idea**
 - ▶ Test: the 2×2 ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**
 - ▶ Test: $\text{QLIKE}(\text{RV}, \text{RV}) = 0$ and $\text{QLIKE}(\text{RV}, F) > 0$ otherwise; the DM variance is HAC with bandwidth at least h

Appendix: The Discovery Loop, Prompts (1/2)

- **Read:** extract exactly what the paper did
 - ▶ “From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed.”
 - ▶ Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map:** what transfers, what must be adapted
 - ▶ “Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it.”
- **Pre-register:** hypotheses, grid, tests, correction, committed before the first result
 - ▶ The assistant drafts; the team decides every number and signs the commit

Appendix: The Discovery Loop, Prompts (2/2)

- **Build and test the pieces:** signatures against identities, not against another library
 - ▶ “Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path.”
- **Run the grid, then attack it**
 - ▶ “Here are all 450 runs (50 assets $\times$ 3 horizons $\times$ 3 specifications: main, depth 2, no return channel). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?”
- **Explain the null:** find the mechanism, not an excuse
 - ▶ Here: flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data

References (1/2)

- ▣ Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536.
- ▣ Barndorff-Nielsen, O.E., Kinnebrock, S., Shephard, N. (2010). Measuring downside risk: realised semivariance. In *Volatility and Time Series Econometrics*, Oxford University Press, 117–136.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society B*, 64(2), 253–280.
- ▣ Belloni, A., Chernozhukov, V. (2013). Least squares after model selection in high-dimensional sparse models. *Bernoulli*, 19(2), 521–547.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 1165–1188.
- ▣ Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18.
- ▣ Chen, K.-T. (1957). Integration of paths, geometric invariants and a generalized Baker–Hausdorff formula. *Annals of Mathematics*, 65(1), 163–178.
- ▣ Chevyrev, I., Kormilitzin, A. (2016). A primer on the signature method in machine learning. [arXiv:1603.03788](https://arxiv.org/abs/1603.03788).
- ▣ Cipollini, F., Cruciani, G., Gallo, G.M., Insana, A., Otranto, E., Spagnolo, F. (2026). VOLatility Archive for Realized Estimates (VOLARE). [arXiv:2602.19732](https://arxiv.org/abs/2602.19732).
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Cuchiero, C., Gazzani, G., Svaluto-Ferro, S. (2023). Signature-based models: theory and calibration. *SIAM Journal on Financial Mathematics*, 14(3), 910–957.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gu, H., Guo, X., Jacobs, T.L., Kaminsky, P., Li, X. (2024). Transportation marketplace rate forecast using signature transform. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4997–5005; [arXiv:2401.04857](https://arxiv.org/abs/2401.04857).

References (2/2)

- Hambly, B., Lyons, T. (2010). Uniqueness for the signature of a path of bounded variation and the reduced path group. *Annals of Mathematics*, 171(1), 109–167.
- Hansen, P.R., Huang, Z., Shek, H.H. (2012). Realized GARCH: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Király, F.J., Oberhauser, H. (2019). Kernels for sequentially ordered data. *Journal of Machine Learning Research*, 20(31), 1–45.
- Levin, D., Lyons, T., Ni, H. (2013). Learning from the past, predicting the statistics for the future, learning an evolving system. arXiv:1309.0260.
- Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- Lyons, T. (1998). Differential equations driven by rough signals. *Revista Matemática Iberoamericana*, 14(2), 215–310.
- Lyons, T., McLeod, A.D. (2022). Signature methods in machine learning. arXiv:2206.14674.
- Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). Nonstandard errors. *Journal of Finance*, 79(3), 2339–2390.
- Morrill, J., Fermanian, A., Kidger, P., Lyons, T. (2020). A generalised signature method for multivariate time series feature extraction. arXiv:2006.00873.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- Patton, A.J., Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3), 683–697.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.
- Salvi, C., Cass, T., Foster, J., Lyons, T., Yang, W. (2021). The signature kernel is the solution of a Goursat PDE. *SIAM Journal on Mathematics of Data Science*, 3(3), 873–899.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society B*, 58(1), 267–288.
- Zou, H. (2006). The adaptive lasso and its oracle properties. *Journal of the American Statistical Association*, 101(476), 1418–1429.